

MAT31 - 91

Probability & Statistics II — Continuous, Limit Theorems & Confidence Intervals

[Module page](#)

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About this textbook

This year, for the first time, every Bachelor course at Albert School has its own textbook, thanks to your teachers and AI.

- The textbook is a help for working on the course at home, after the class.
- The course as taught in class remains the reference, and prevails over the textbook wherever the two differ.
- This is a first version. Help us improve it by sharing your feedback with your teacher.

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Preface

By the end of this book you will be able to work with **continuous** random variables — those that take any value in an interval, such as a waiting time, a height, or a measurement error — with the fluency you already have for counts. Concretely, you will be able to:

- evaluate the improper integrals that continuous densities require, and tell which of them converge;
- describe a continuous variable by its density and its cumulative distribution function, read probabilities as areas, and compute its expectation and variance;
- compute probabilities under the uniform, exponential and normal laws — by standardizing and reading a normal table — and recognise the chi-squared and Student- t laws as the tools for a normal sample's variance and small-sample mean;
- state the Law of Large Numbers and the Central Limit Theorem, tell the spread of a single observation, σ , from the spread of an average, $\sigma/\sqrt{n}$, and use the normal approximation of the binomial;
- bound a tail probability with Chebyshev's inequality, assuming nothing about the shape of the distribution;
- estimate an expectation by Monte Carlo simulation and attach an honest confidence interval to the estimate;
- build a confidence interval for a mean and read it correctly, and carry out a hypothesis test — choosing a z - or t -test, handling one- and two-sided p -values, and avoiding the fallacies that surround both.

These are the tools on which reported error bars, A/B tests, and every claim of statistical significance rest.

To follow it you need two things this book takes for granted and no chapter re-teaches. From *Foundations* you need to integrate fluently — the fundamental theorem of calculus, substitution, integration by parts, and Chasles' relation — because a continuous variable is defined and computed entirely through integrals. From *Probability & Statistics I (discrete)* you need discrete probability — random variables, expectation and variance, and the classic discrete laws — because the continuous theory is built here as its mirror image, and almost every idea has a discrete ancestor we point back to.

1. Generalized integrals

A continuous random variable will spread its probability over an interval, and total probability must equal 1. When that interval is the whole real line — as for the normal distribution — computing “total probability” means integrating over an **unbounded** region. Ordinary definite integrals $\int_a^b$ run between two finite numbers; we first need to make sense of integrals that run to infinity or whose integrand grows without bound. These are **generalized** (or **improper**) integrals.

We treat them by intuition and a few standard reference cases, not by the full convergence theory. One idea guides everything: **an improper integral is a limit of ordinary ones.**

1.1. Integrating to infinity

Example — the area under a decaying tail. Consider $f(x) = 1/x^2$ for $x \geq 1$. Its graph comes ever closer to the axis as x grows, but never touches it — the region under it stretches infinitely far to the right. Does it nonetheless enclose a **finite** area? Integrate up to a finite cutoff b and observe:

$$\int_1^b \frac{1}{x^2} dx = \left[-\frac{1}{x} \right]_1^b = 1 - \frac{1}{b}.$$

As $b \rightarrow \infty$ the term $1/b$ vanishes, leaving area exactly 1. An infinitely long region can have finite area.

Definition 1 (improper integral on an unbounded interval). If $\int_a^b f(x) dx$ exists for every $b > a$, we define

$$\int_a^\infty f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx.$$

When the limit exists and is finite, the integral **converges**; otherwise it **diverges**. Integrals to $-\infty$, and the doubly-infinite $\int_{-\infty}^\infty f = \int_{-\infty}^c f + \int_c^\infty f$, are defined the same way.

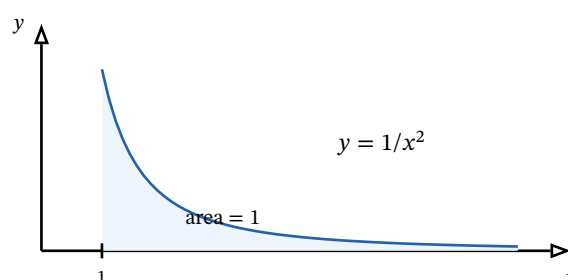


Figure 1: The improper integral $\int_1^\infty x^{-2} dx$. The region is unbounded to the right yet has finite area 1, because the tail thins fast enough. We compute it as the limit of the finite areas $1 - 1/b$ as the cutoff $b \rightarrow \infty$.

The decisive question is **how fast the tail decays**. Compare $1/x^2$ with $1/x$:

$$\int_1^b \frac{1}{x} dx = [\ln x]_1^b = \ln b \rightarrow \infty.$$

The area under $1/x$ is **infinite** — the curve decays too slowly. This is the canonical reference family:

Proposition 1 (the p -integral at infinity).

$$\int_1^\infty \frac{1}{x^p} dx \text{ converges} \iff p > 1, \text{ and then equals } \frac{1}{p-1}.$$

Proof. For $p \neq 1$, $\int_1^b x^{-p} dx = [x^{1-p}/(1-p)]_1^b = (b^{1-p} - 1)/(1-p)$. If $p > 1$ the exponent $1-p$ is negative, so $b^{1-p} \rightarrow 0$ and the limit is $1/(p-1)$. If $p < 1$ the exponent is positive and $b^{1-p} \rightarrow \infty$. The borderline $p = 1$ gives $\ln b \rightarrow \infty$ as above. $\square$

A second indispensable case is the **exponential tail**, which decays faster than any power and will underlie the exponential and normal distributions:

$$\int_0^\infty e^{-\lambda x} dx = \left[-\frac{1}{\lambda} e^{-\lambda x} \right]_0^\infty = \frac{1}{\lambda} \quad (\lambda > 0).$$

1.2. Integrating an unbounded function

The other way an integral turns improper is when the **integrand** grows without bound at an endpoint, even over a finite interval. The remedy is the same — stop short at a cutoff and take a limit.

Example — a spike at the origin. The function $1/\sqrt{x}$ is unbounded as $x \rightarrow 0^+$. Yet

$$\int_0^1 \frac{1}{\sqrt{x}} dx = \lim_{a \rightarrow 0^+} \int_a^1 x^{-1/2} dx = \lim_{a \rightarrow 0^+} \left[2\sqrt{x} \right]_a^1 = \lim_{a \rightarrow 0^+} (2 - 2\sqrt{a}) = 2.$$

The spike is integrable: its area is finite. By contrast $\int_0^1 (1/x) dx = \lim_{a \rightarrow 0^+} (-\ln a) = +\infty$ diverges.

Notice the mirror image of the tail case: near 0, $1/x^p$ converges exactly when $p < 1$ — the opposite inequality. The intuition: at infinity you need **fast decay** (p large); at a spike you need a **mild singularity** (p small).

Pitfall. Do not evaluate an improper integral by blindly plugging in the limits as if they were ordinary numbers. Writing “ $[-1/x]_1^\infty = 0 - (-1)$ ” gives the right answer here only because the limit happens to exist; with $\int_1^\infty (1/x) dx$ the same reflex hides a divergence. **Always** set up the limit explicitly when an endpoint is ∞ or a point where f is undefined.

Remark. A useful shortcut when you only need to know **whether** an integral converges, not its value, is **comparison**: if $0 \leq f(x) \leq g(x)$ and $\int g$ converges, so does $\int f$. For instance $\int_1^\infty e^{-x^2} dx$ converges because $e^{-x^2} \leq e^{-x}$ for $x \geq 1$ and the exponential integral converges — even though

e^{-x^2} has no elementary antiderivative. This is exactly why the normal distribution's normalising constant has to be found by a trick rather than by direct integration.

2. Continuous random variables

A discrete variable places lumps of probability on isolated points. A **continuous** variable smears probability over an interval, so that any **single** value has probability zero and only **ranges** carry weight. The lump-heights $p_X(x)$ are replaced by a **density** $f_X(x)$, and every sum $\sum_x$ becomes an integral $\int dx$. This one substitution carries almost all of the discrete theory across.

2.1. Density and cumulative distribution function

Example — a random point on a stick. Snap a 1-metre stick at a uniformly random point and let X be the distance of the break from the left end. Then X lands anywhere in $[0, 1]$ with no preference. What is $\mathbb{P}(X = 0.5)$ — exactly the midpoint? Zero: there are infinitely many points and no single one can carry positive probability. But $\mathbb{P}(0.4 \leq X \leq 0.6) = 0.2$, the **length** of the sub-interval. Probability has become **length**, and more generally **area under a curve**.

Definition 2 (probability density function). A **continuous random variable** X is described by a **probability density function** (PDF) $f_X(x) \geq 0$ such that probabilities are **areas**:

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x) dx,$$

and the total area is one, $\int_{-\infty}^{\infty} f_X(x) dx = 1$.

The density is **not** a probability — it can exceed 1 (a sharp spike over a short interval). Only its **integral** over a range is a probability. The correct infinitesimal reading is $\mathbb{P}(x \leq X \leq x + dx) \approx f_X(x) dx$: the density times a tiny width gives the probability in that sliver.

Pitfall. Because single points have probability zero, the endpoints never matter for a continuous variable: $\mathbb{P}(a \leq X \leq b) = \mathbb{P}(a < X < b)$. Strict and non-strict inequalities give the same answer — a relief after the fussy off-by-one bookkeeping of discrete sums. (This fails completely for discrete variables, where $\mathbb{P}(X \leq 3)$ and $\mathbb{P}(X < 3)$ can differ by $p_X(3)$.)

As in the discrete case, the **cumulative distribution function** accumulates probability from the left — but now by integrating, not summing.

Definition 3 (cumulative distribution function). The **CDF** of X is $F_X(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^x f_X(t) dt$. It is non-decreasing, runs from $F_X(-\infty) = 0$ to $F_X(+\infty) = 1$, and — by the Fundamental Theorem of Calculus — recovers the density by differentiation:

$$(F_X)'(x) = f_X(x).$$

This last identity is the through-line of the whole chapter: **density and CDF are a derivative–integral pair**, exactly the Fundamental Theorem of Calculus you already know. To get probabilities from a density, integrate; to get the density from a CDF, differentiate; and $\mathbb{P}(a \leq X \leq b) = F_X(b) - F_X(a)$.

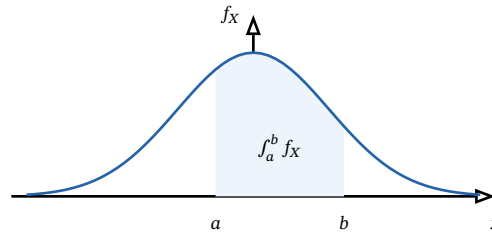


Figure 2: Probability as area. The shaded region under the density f_X between a and b equals $\mathbb{P}(a \leq X \leq b)$. The whole region under the curve has area 1. Individual points, being slivers of zero width, carry zero probability.

2.2. Expectation, variance, and moments

Every definition from the discrete theory transfers by turning $\sum_x x p_X(x)$ into $\int x f_X(x) dx$. The **interpretation is identical** — expectation is the balance point of the density, variance its spread — only the computation changes from summing to integrating.

Definition 4 (expectation and variance). For a continuous variable X with density f_X ,

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x f_X(x) dx, \quad \mathbb{V}(X) = \mathbb{E}((X - \mu)^2),$$

where $\mu = \mathbb{E}(X)$, provided the integrals converge absolutely. The k -th **moment** is $\mathbb{E}(X^k) = \int x^k f_X(x) dx$.

Expanding the square gives the **König–Huygens formula**, the shortcut you will actually compute with — it trades the awkward $\mathbb{E}((X - \mu)^2)$ for the two plain moments $\mathbb{E}(X^2)$ and $\mathbb{E}(X)$.

Proposition 2 (König–Huygens formula).

$$\mathbb{V}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2.$$

Proof. Start from the variance as an expected squared deviation, write $\mu = \mathbb{E}(X)$, expand the square, and use linearity of expectation (which the integral inherits from the sum):

$$\begin{aligned} \mathbb{V}(X) &= \mathbb{E}((X - \mu)^2) \\ &= \mathbb{E}(X^2 - 2\mu X + \mu^2) \\ &= \mathbb{E}(X^2) - 2\mu \mathbb{E}(X) + \mu^2. \end{aligned}$$

Now replace $\mathbb{E}(X)$ by the mean μ it names: the last two terms collapse to $-2\mu^2 + \mu^2 = -\mu^2$. Substituting back leaves $\mathbb{V}(X) = \mathbb{E}(X^2) - \mu^2$. The identity is purely algebraic — it holds for discrete and continuous variables alike, because both expectations are linear. $\square$

The **transfer theorem** (the “law of the unconscious statistician”) also survives verbatim, and it is what lets us compute $\mathbb{E}(X^2)$ for the variance without first finding the distribution of X^2 .

Theorem 1 (Transfer theorem). For any (measurable) function g ,

$$\mathbb{E}(g(X)) = \int_{-\infty}^{\infty} g(x) f_X(x) dx.$$

Linearity of expectation, $\mathbb{V}(aX + b) = a^2\mathbb{V}(X)$, and variance-additivity for independent variables all hold word-for-word as in the discrete case — their proofs used only linearity of the sum, which is shared by the integral.

Example — mean and variance of a point on the stick. For the uniform break point, the density is $f_X(x) = 1$ on $[0, 1]$ and 0 elsewhere (it must integrate to 1 over a unit interval). Then

$$\mathbb{E}(X) = \int_0^1 x \cdot 1 dx = \left[\frac{x^2}{2} \right]_0^1 = \frac{1}{2}, \quad \mathbb{E}(X^2) = \int_0^1 x^2 dx = \frac{1}{3},$$

so $\mathbb{V}(X) = 1/3 - (1/2)^2 = 1/12$. The mean sits at the centre, as symmetry demands, and the standard deviation $\sigma = 1/\sqrt{12} \approx 0.29$ measures the typical distance from it.

Pitfall. A continuous variable can have **no** mean. If the tails of f_X are too heavy the defining integral diverges — the Cauchy density $f(x) = 1/(\pi(1 + x^2))$ is the classic example, with $\int x f(x) dx$ divergent. “Expectation exists” is a genuine condition, not a formality, exactly as absolute convergence was for discrete sums.

3. Classic continuous distributions

Just as four discrete families covered most discrete phenomena, a small catalogue of continuous distributions covers most continuous ones. We derive each from its **modelling assumption**, exactly the discipline of the discrete theory: the formula should feel **inevitable**, not memorised.

3.1. The uniform distribution

Definition 5 (uniform distribution). $X \sim \text{Unif}(a, b)$ is “no preference within $[a, b]$ ”: its density is constant,

$$f_X(x) = \frac{1}{b-a} \quad \text{for } a \leq x \leq b, \quad 0 \text{ otherwise.}$$

Then $\mathbb{E}(X) = (a+b)/2$ (the midpoint) and $\mathbb{V}(X) = (b-a)^2/12$.

The constant height $1/(b-a)$ is forced by total area 1. The uniform is the continuous analogue of equiprobability, and — crucially for the last chapter — it is the output of a standard random-number generator, the raw material from which every other distribution will be simulated.

3.2. The exponential distribution and memorylessness

Example — waiting for the next bus. Buses arrive “at random” at an average rate of λ per hour, with no schedule — how long you have already waited tells you nothing about how long is left. Let T be the time you wait. Slice the wait into tiny steps of length h ; in each step a bus either comes or not, independently, with probability λh (rate $\times$ duration). Waiting longer than t then means no bus in any of its $n = t/h$ steps — a geometric waiting time — so

$$\mathbb{P}(T > t) = (1 - \lambda h)^{t/h} \rightarrow e^{-\lambda t} \quad \text{as } h \rightarrow 0,$$

using $(1 - \lambda h)^{1/h} \rightarrow e^{-\lambda}$. Hence the survival function is $\mathbb{P}(T > t) = e^{-\lambda t}$, the CDF is $F_T(t) = 1 - e^{-\lambda t}$, and differentiating gives the density.

Definition 6 (exponential distribution). $T \sim \text{Exp}(\lambda)$ models the waiting time for an event that occurs at random with a constant average rate $\lambda > 0$:

$$f_T(t) = \lambda e^{-\lambda t}, \quad F_T(t) = 1 - e^{-\lambda t} \quad (t \geq 0).$$

Its mean and standard deviation are both $1/\lambda$: $\mathbb{E}(T) = 1/\lambda$ and $\mathbb{V}(T) = 1/\lambda^2$.

The mean $1/\lambda$ is intuitive: at rate λ events per hour you wait on average $1/\lambda$ hours — the continuous echo of the geometric mean $1/p$. Indeed the exponential is the **continuous limit of the geometric**, and it inherits the geometric’s defining quirk.

Theorem 2 (Memorylessness of the exponential). The exponential distribution is **memoryless**:

$$\mathbb{P}(T > s + t \mid T > s) = \mathbb{P}(T > t) \quad \text{for all } s, t \geq 0.$$

Proof. The survival function is $G(t) = \mathbb{P}(T > t) = e^{-\lambda t}$, from the example above. By the definition of conditional probability,

$$\mathbb{P}(T > s + t \mid T > s) = \frac{\mathbb{P}(T > s + t \text{ and } T > s)}{\mathbb{P}(T > s)}.$$

The event $\{T > s + t\}$ is contained in $\{T > s\}$, so their intersection is $\{T > s + t\}$ itself and the numerator is simply $\mathbb{P}(T > s + t)$. Substituting the survival function and simplifying the exponentials,

$$\mathbb{P}(T > s + t \mid T > s) = \frac{\mathbb{P}(T > s + t)}{\mathbb{P}(T > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = \mathbb{P}(T > t).$$

The exponential turns the sum $s + t$ in the exponent into a product, which is exactly the factor the conditioning divides away. $\square$

Remark (and it is the only memoryless continuous law). The converse holds too, so memorylessness **characterizes** the exponential. Write the survival function $G(t) = \mathbb{P}(T > t)$. The memoryless condition is exactly the multiplicative equation

$$G(s + t) = G(s)G(t) \quad \text{for all } s, t \geq 0,$$

and the only decreasing solutions with $G(0) = 1$ are the exponentials $G(t) = e^{-\lambda t}$ (a sum in the argument becoming a product in the value forces this form, exactly as $2^{s+t} = 2^s 2^t$). Hence **any** memoryless continuous lifetime on $[0, \infty)$ must be $\text{Exp}(\lambda)$. We use only the forward direction above; the converse is what lets us say the exponential — and nothing else — models “no ageing”.

Remark (the exponential is the geometric's continuous limit). The derivation of the bus example is exactly the geometric law refined to continuous time. The geometric is the **discrete** memoryless law — waiting for the first success in independent trials, with $\mathbb{P}(N > n) = (1 - p)^n$ — and shrinking the trial length h turns its $(1 - p)^n$ into the exponential's $e^{-\lambda t}$: the **same** memoryless mechanism, discrete trials refined into continuous time. This is why the exponential's mean $1/\lambda$ mirrors the geometric mean $1/p$.

Pitfall. Memorylessness is deeply counter-intuitive and a common source of error. A lightbulb with exponential lifetime that has already worked for 1000 hours has, **probabilistically**, the same remaining life as a brand-new bulb: its future lifetime has the same distribution as a fresh bulb's. This is a **property of the model**, appropriate for things that fail by sudden random shocks (radioactive decay, memoryless queues) but wrong for things that **wear out** (human lifespans, mechanical fatigue), whose failure rate rises with age. Choosing the exponential asserts “no ageing”.

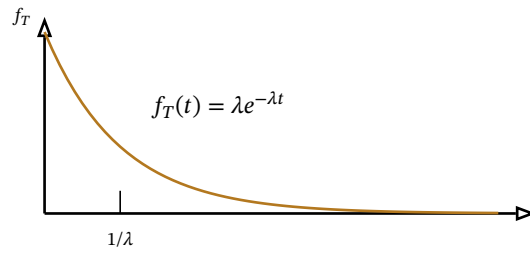


Figure 3: The exponential density for $\lambda = 1$. It is highest at $t = 0$ and decays at a constant relative rate — the geometric signature of “no ageing”. The mean $1/\lambda$ lies to the right of the peak because the long right tail pulls the balance point outward.

3.3. The normal distribution

The normal (Gaussian) distribution is the centrepiece of the course. It will emerge **inevitably** from the Central Limit Theorem in Chapter 5; here we meet its shape and learn to compute with it.

Example — the spread of adult heights. Record the heights of a large group of adults and plot how they cluster. The histogram is a single smooth hump: symmetric about a central value, with most people close to it and steadily fewer as you move out, falling away at the same rate on each side — no second peak, no long tail on one side only. Countless quantities scatter this way: repeated readings of one fixed length blurred by measurement error, or any total built from many small independent effects. This recurring bell shape is the **normal distribution**, and the definition below gives the exact curve that models it.

Definition 7 (normal distribution). $X \sim \mathcal{N}(\mu, \sigma^2)$ has the bell-shaped density

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

Its parameters are exactly its mean and variance: $\mathbb{E}(X) = \mu$, $\mathbb{V}(X) = \sigma^2$. The curve is symmetric about μ , with inflection points at $\mu \pm \sigma$.

The two parameters do the obvious geometric jobs: μ **locates** the peak, σ sets the **width**. The prefactor $1/(\sigma\sqrt{2\pi})$ is the normalising constant that makes the total area 1 — and as noted in Chapter 1, the integral $\int e^{-x^2/2} dx = \sqrt{2\pi}$ cannot be found by an elementary antiderivative. We meet it, and the variations of it you will need to compute, in Section 3.3.3 below.

3.3.1. Standardization

Because no antiderivative exists, normal probabilities are read from a **table** (or a calculator). We cannot tabulate every (μ, σ) , so we reduce **every** normal to a single one by **standardizing**.

Theorem 3 (Standardization). If $X \sim \mathcal{N}(\mu, \sigma^2)$, then

$$Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1),$$

the **standard normal**. Consequently

$$\mathbb{P}(X \leq x) = \mathbb{P}\left(Z \leq \frac{x - \mu}{\sigma}\right) = \Phi\left(\frac{x - \mu}{\sigma}\right),$$

where Φ is the standard normal CDF, the tabulated function.

Subtracting μ re-centres the variable at 0; dividing by σ rescales it to unit spread. The transformed value $z = (x - \mu)/\sigma$ — the **z-score** — answers “how many standard deviations above the mean is x ?”, a unit-free question whose answer is the same for every normal.

Example — test scores. Exam marks are $\mathcal{N}(\mu = 500, \sigma^2 = 100^2)$. What fraction of candidates score above 650? Standardize: $z = (650 - 500)/100 = 1.5$. So $\mathbb{P}(X > 650) = \mathbb{P}(Z > 1.5) = 1 - \Phi(1.5) \approx 1 - 0.9332 = 0.0668$ — about 6.7%. The raw mark 650 became the universal statement “1.5 standard deviations above average”, and the table supplied the probability.

The following table gives $\Phi(z) = \mathbb{P}(Z \leq z)$ for the standard normal. Read it by splitting z into its first two digits (the row) and its second decimal (the column): for example $\Phi(1.96)$ sits in row 1.9, column .06, giving 0.9750.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

For **negative** z , symmetry gives $\Phi(-z) = 1 - \Phi(z)$, so the table need only list $z \geq 0$. Two values are worth memorising because they recur in every confidence interval: $\Phi(1.645) \approx 0.95$ and $\Phi(1.96) \approx 0.975$.

Pitfall. The table gives $\Phi(z) = \mathbb{P}(Z \leq z)$, the area to the **left**. For a **right** tail use $\mathbb{P}(Z > z) = 1 - \Phi(z)$; for a **two-sided** range use $\mathbb{P}(-z \leq Z \leq z) = 2\Phi(z) - 1$; and for **negative** arguments use $\Phi(-z) = 1 - \Phi(z)$. Forgetting which tail you are in is the single most common error in this chapter — always sketch the bell curve and shade the region you want before reaching for a number.

3.3.2. The 68–95–99.7 rule

Proposition 3 (the empirical rule). For any normal variable,

$$\begin{aligned}\mathbb{P}(\mu - \sigma \leq X \leq \mu + \sigma) &\approx 68\%, \\ \mathbb{P}(\mu - 2\sigma \leq X \leq \mu + 2\sigma) &\approx 95\%, \\ \mathbb{P}(\mu - 3\sigma \leq X \leq \mu + 3\sigma) &\approx 99.7\%.\end{aligned}$$

These follow directly from the table: $2\Phi(1) - 1 \approx 0.68$, $2\Phi(2) - 1 \approx 0.954$, $2\Phi(3) - 1 \approx 0.997$. The rule is the fastest sanity-check in statistics: a normal value almost never strays beyond three standard deviations, and a value two deviations out is already a one-in-twenty event.

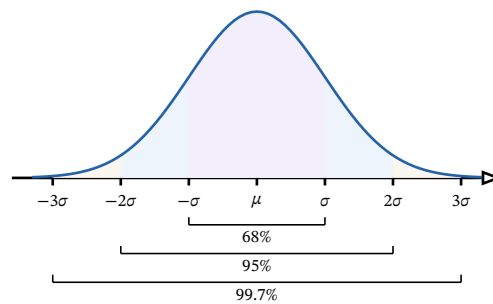


Figure 4: The 68–95–99.7 rule. The three brackets below the axis span one, two and three standard deviations either side of the mean; the percentage on each is the **total** mass inside that span — about 68%, 95% and 99.7%. A value beyond 3σ is genuinely rare.

3.3.3. The Gaussian integral and its variations

Because the bell curve has no elementary antiderivative, the very fact that it is a density — that it integrates to 1 — rests on one remarkable definite integral.

Theorem 4 (the Gaussian integral).

$$\int_{-\infty}^{\infty} e^{-x^2/2} dx = \sqrt{2\pi}.$$

With no antiderivative to evaluate, the usual method is unavailable. The trick is to compute the **square** of the integral as a double integral and pass to polar coordinates — we sketch it as motivation only; the details are not examinable. Writing I for the integral and using a dummy variable y for the second copy,

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2/2} dx \right) \left(\int_{-\infty}^{\infty} e^{-y^2/2} dy \right) = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)/2} dx dy.$$

In polar coordinates $x^2 + y^2 = r^2$ and the area element is $dx dy = r dr d\theta$, so the integrand becomes $e^{-r^2/2}r$ — which **does** have an elementary antiderivative, $-e^{-r^2/2}$:

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2/2} r dr d\theta = \int_0^{2\pi} 1 d\theta = 2\pi,$$

hence $I = \sqrt{2\pi}$. The extra factor r from the area element is exactly what makes the radial integral elementary — the whole trick in one line.

What you **do** need to compute fluently are **variations** of this integral, reached by two standard moves that both reduce back to the value $\sqrt{2\pi}$.

Method 1 (scaling). For a Gaussian with a width parameter, substitute $x = \sigma u$ (so $dx = \sigma du$):

$$\int_{-\infty}^{\infty} e^{-x^2/(2\sigma^2)} dx = \sigma \int_{-\infty}^{\infty} e^{-u^2/2} du = \sigma\sqrt{2\pi}.$$

This is exactly the calculation the normalising constant undoes: dividing by $\sigma\sqrt{2\pi}$ makes the $\mathcal{N}(\mu, \sigma^2)$ density integrate to 1. In the bare form $\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\pi/a}$ for any $a > 0$ (the case $a = 1/(2\sigma^2)$).

Method 2 (completing the square). For a linear term in the exponent, complete the square. For $a > 0$,

$$-ax^2 + bx = -a\left(x - \frac{b}{2a}\right)^2 + \frac{b^2}{4a},$$

so, substituting $u = x - b/(2a)$ and using the scaling result,

$$\int_{-\infty}^{\infty} e^{-ax^2+bx} dx = e^{b^2/(4a)} \int_{-\infty}^{\infty} e^{-au^2} du = \sqrt{\frac{\pi}{a}} e^{b^2/(4a)}.$$

Example — a shifted Gaussian. Compute $\int_{-\infty}^{\infty} e^{-x^2+4x} dx$. Here $a = 1$ and $b = 4$, so $-x^2 + 4x = -(x - 2)^2 + 4$ and

$$\int_{-\infty}^{\infty} e^{-x^2+4x} dx = e^4 \int_{-\infty}^{\infty} e^{-(x-2)^2} dx = e^4\sqrt{\pi} \approx 96.8.$$

The shift $x - 2$ relocates the peak but leaves the area unchanged; the factor e^4 is the height the completed square pulls outside the integral.

3.4. Two distributions for inference: chi-squared and Student- t

The last two families exist to **support statistical inference** — they describe how sample statistics behave. We meet them briefly here; the Student- t returns in Chapter 8, where it replaces the normal for confidence intervals built from a small sample.

Example — how a sample's spread behaves. Draw a handful of readings from a normal population and measure how far they scatter, by adding up their squared distances from the centre. Do it again on a fresh sample and this total comes out different every time: never negative, usually moderate, now and then large. Its distribution is therefore bunched against zero with a long tail to the right — nothing like a symmetric bell. That right-skewed shape, arising whenever we sum squared normal deviations, is what the **chi-squared** family below describes.

Definition 8 (chi-squared distribution). If $Z_1, \dots, Z_k$ are independent standard normals, then

$$\chi_k^2 = Z_1^2 + \dots + Z_k^2$$

has the **chi-squared distribution with k degrees of freedom**. It lives on $[0, \infty)$, has mean k and variance $2k$, and is right-skewed.

Being a sum of squares, χ^2 is the natural distribution of **sample variances** and of goodness-of-fit discrepancies — the basis of tests you will meet in later statistics courses. As k grows it becomes more symmetric, a foretaste of the Central Limit Theorem acting on the squared normals.

Example — locating a mean from a few readings. Suppose you have just eight measurements and want to pin down their true mean, but you must estimate the spread from those same eight numbers. Standardizing with the normal law would pretend the spread is known exactly; in reality your estimate of it wobbles, and on so small a sample that wobble makes large standardized values turn up more often than the normal predicts. The bell-shaped law with exactly these fatter tails — the price of not knowing the spread — is the **Student- t** below.

Definition 9 (Student- t distribution). If $Z \sim \mathcal{N}(0, 1)$ and χ_k^2 are independent, then

$$T = \frac{Z}{\sqrt{\chi_k^2/k}}$$

has the **Student- t distribution with k degrees of freedom**. It is symmetric and bell-shaped like the normal but with **heavier tails**; as $k \rightarrow \infty$ it converges to $\mathcal{N}(0, 1)$.

The t -distribution is what we must use in place of the normal when the population variance is **unknown** and estimated from a **small** sample: the extra uncertainty in estimating σ fattens the tails, so t produces slightly wider — more honest — confidence intervals. For large samples the difference vanishes and the normal returns.

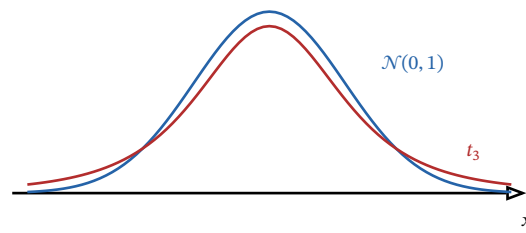


Figure 5: Standard normal (blue) versus Student- t with 3 degrees of freedom (red). Both are symmetric and bell-shaped, but t has a lower peak and **heavier tails**: extreme values are more probable, reflecting the extra uncertainty of a small sample. As the degrees of freedom grow, t tightens onto the normal.

4. The Law of Large Numbers

We now reach the theorems that justify the entire practice of statistics. The first answers the most basic question one could ask of probability: **if I repeat an experiment many times, does the average outcome settle down to the expected value?**

4.1. Two tail inequalities: Markov and Chebyshev

Before the theorem itself we need two inequalities that bound how much probability a variable can place far from its mean — using **only** its mean and variance, whatever its distribution. They are the engine of the proof, and useful in their own right whenever the exact distribution is unknown.

Theorem 5 (Markov's inequality). If $X \geq 0$ and $a > 0$, then

$$\mathbb{P}(X \geq a) \leq \mathbb{E} \frac{X}{a}.$$

Proof. Because $X \geq 0$, its expectation is the improper integral $\mathbb{E}(X) = \int_0^\infty x f_X(x) dx$ of Chapter 1. Split the range at a by Chasles' relation, discard the first piece — non-negative, since its integrand is — and on the second, where $x \geq a$, bound the integrand below by $a f_X(x)$:

$$\begin{aligned} \mathbb{E}(X) &= \int_0^a x f_X(x) dx + \int_a^\infty x f_X(x) dx \\ &\geq \int_a^\infty x f_X(x) dx \\ &\geq \int_a^\infty a f_X(x) dx = a \int_a^\infty f_X(x) dx = a \mathbb{P}(X \geq a). \end{aligned}$$

Dividing by $a > 0$ gives the claim. □

Markov uses only the mean, so it is crude; feeding it the **squared deviation** $(X - \mu)^2$ — a non-negative variable whose mean is precisely the variance — sharpens it into the bound every later result leans on.

Theorem 6 (Chebyshev's inequality). For any X with mean μ and variance σ^2 , and any $\varepsilon > 0$,

$$\mathbb{P}(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}.$$

Proof. Apply Markov's inequality to the non-negative variable $(X - \mu)^2$ with cutoff $a = \varepsilon^2$. The event $|X - \mu| \geq \varepsilon$ is identical to $(X - \mu)^2 \geq \varepsilon^2$, and $\mathbb{E}((X - \mu)^2) = \mathbb{V}(X) = \sigma^2$, so

$$\mathbb{P}(|X - \mu| \geq \varepsilon) = \mathbb{P}((X - \mu)^2 \geq \varepsilon^2) \leq \mathbb{E} \frac{(X - \mu)^2}{\varepsilon^2} = \frac{\sigma^2}{\varepsilon^2}.$$

□

Example — a distribution-free tail bound. A delivery time has mean $\mu = 100$ minutes and standard deviation $\sigma = 15$, but we know **nothing** about its shape. What can we say about being more than 30 minutes off? With $\varepsilon = 30 = 2\sigma$,

$$\mathbb{P}(|X - 100| \geq 30) \leq \frac{15^2}{30^2} = \frac{1}{4},$$

so at least 75% of deliveries land in $[70, 130]$ — for **any** distribution with this mean and variance. If we knew X were normal, the empirical rule would tighten this to about 95%; Chebyshev is the price of assuming nothing.

Pitfall. Chebyshev’s bound is deliberately loose: it must hold for **every** distribution, so it is far from tight for any particular one. Read “ $\mathbb{P}(|X - \mu| \geq 2\sigma) \leq 1/4$ ” as a **guarantee**, never an estimate — the true probability is usually far smaller (about 0.05 for a normal). Its power is exactly that it needs no distributional assumption.

4.2. Statement and interpretation

Example — the running average of coin flips. Flip a fair coin and record the **running proportion** of heads after 1, 2, 3, ... flips. Early on it jumps around — 0, then 0.5, then 0.33 — but as flips accumulate it stops moving and settles near 0.5. Not because some force “corrects” imbalances, but because each new flip changes an ever-larger total by an ever-smaller amount. The average **stabilises on its own**.

Theorem 7 (Weak law of large numbers). Let $X_1, X_2, \dots$ be independent and identically distributed with mean μ and finite variance. Their **sample mean**

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

converges to μ as $n \rightarrow \infty$: for any tolerance $\varepsilon > 0$,

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \rightarrow 0.$$

Proof. Compute the two moments of the sample mean: by linearity $\mathbb{E}(\bar{X}_n) = \mu$, and by variance-additivity for independent variables $\mathbb{V}(\bar{X}_n) = \sigma^2/n$. So $\bar{X}_n$ is an unbiased estimator of μ whose spread collapses as n grows. Apply Chebyshev’s inequality to $\bar{X}_n$ and let $n \rightarrow \infty$ in the bound:

$$0 \leq \mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \mathbb{V} \frac{\bar{X}_n}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0.$$

The probability is squeezed between 0 and a bound that vanishes, so it tends to 0 itself. That is the whole theorem: the variance of an average falls like $1/n$, and Chebyshev turns that into convergence. $\square$

The proof reveals **why** it works and connects straight back to the discrete case: the variance of an average of n independent copies is $1/n$ of a single one’s, so spread shrinks like $1/n$ — the same $p(1 -$

$p)/n$ we found for the sample proportion. The LLN is the formal promise that **more data means a more reliable average**.

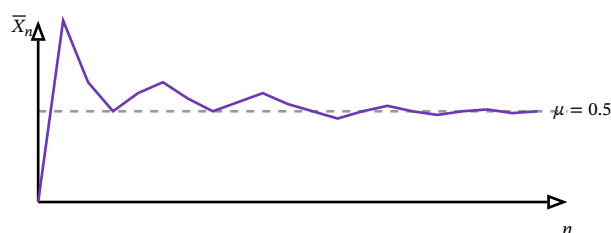


Figure 6: The Law of Large Numbers in action: the running proportion of heads swings wildly at first, then converges to $\mu = 0.5$. The amplitude of the swings shrinks like $1/\sqrt{n}$ — the convergence is real but gradual.

The theorem above is the **weak** law: for each fixed tolerance ε , the chance of the sample mean being off by more than ε tends to 0. A sharper statement, the **strong** law, is also true and worth knowing.

Theorem 8 (Strong law of large numbers). Under the same hypotheses (i.i.d. with mean μ), the sample mean converges to μ **almost surely**:

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} \bar{X}_n = \mu\right) = 1.$$

The two laws differ in their **mode of convergence**, and the distinction is the technical content here. The weak law is **convergence in probability**: it pins down, at each single large n , the probability of a gap — but leaves open that a rare excursion still surfaces now and then as n marches on. The strong law is **almost-sure convergence**: it speaks of the entire trajectory at once, promising that with probability 1 the running average eventually settles onto μ and never leaves again — only finitely many excursions ever occur. The strong law implies the weak, so it is the stronger guarantee; we prove only the weak one, whose Chebyshev argument is elementary, and state the strong one without proof. For everyday statistics both underwrite the same conclusion — **more data means a more trustworthy average** — which is why the weak law’s simple, quotable bound $\sigma^2/(n\varepsilon^2)$ is the one we actually reach for.

4.3. The Gambler’s Fallacy

Pitfall. The LLN says the **average** converges; it does **not** say that nature keeps a running account and pays back past deficits. After a run of 10 heads, the **Gambler’s Fallacy** insists tails is now “due”. It is not: the coin has no memory, so the next flip is still 50/50. Convergence happens by **dilution**, not **compensation** — the early excess of heads is not cancelled by future tails; it is simply outweighed by the growing denominator n . In fact the absolute imbalance (heads minus tails) can keep **growing** even as the proportion converges. Confusing “the proportion tends to $1/2$ ” with “the counts must even out” is one of the most consequential errors in all of probability.

4.4. Implication for sampling

The LLN is the licence to **estimate by sampling**. The sample proportion of voters favouring a candidate converges to the true proportion; the sample mean income converges to the population

mean. It guarantees that the estimate is **eventually** right — but it is silent on **how fast**, and on **how wrong** a finite sample might be. For that quantitative question we need the next theorem.

5. The Central Limit Theorem

The LLN tells us the sample mean lands on μ ; the **Central Limit Theorem** tells us **how it is distributed around μ** on the way there. Its answer is astonishing and universal: **whatever the shape of the original distribution, the sample mean is approximately normal**. This is why the bell curve is everywhere.

5.1. Statement

Example — building a bell from dice. A single die is uniform on $1, \dots, 6$ — flat, nothing like a bell. Add a second die: the **sum** runs over $2, \dots, 12$ and its distribution is already triangular, peaked at 7. Add more, and by ten dice the histogram of the sum is a smooth bell (Figure 7). The flat shape of the individual die **disappears**; only its mean and variance carry into the shape of the result. The sample **average** is this same sum divided by n , so it inherits the very same bell shape — but centred on the fixed mean 3.5 and, as the standard error $\sigma/\sqrt{n}$ below makes precise, growing ever **narrower** around it as n grows. Summing — or averaging — many independent quantities **produces** normality.

Theorem 9 (Central Limit Theorem). Let $X_1, \dots, X_n$ be i.i.d. with mean μ and finite variance σ^2 . As $n \rightarrow \infty$, the standardized sample mean converges in distribution to a standard normal:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{\mathcal{L}} \mathcal{N}(0, 1).$$

The label $\mathcal{L}$ on the arrow marks **convergence in distribution** — the mode the Central Limit Theorem asserts — distinct from the convergence in probability of the weak law and the almost-sure convergence of the strong law in Chapter 4. Equivalently, for large n , $\bar{X}_n \approx \mathcal{N}(\mu, \sigma^2/n)$ and the sum $S_n = \sum X_i \approx \mathcal{N}(n\mu, n\sigma^2)$.

The quantity $\sigma/\sqrt{n}$ — the standard deviation of the sample mean — is called the **standard error**. The $\sqrt{n}$ governs the whole of sampling: to halve the error you need **four times** the data. The LLN said the error $\rightarrow 0$; the CLT says it does so at rate $1/\sqrt{n}$ and with a **normal** shape, which is exactly what makes confidence intervals computable.

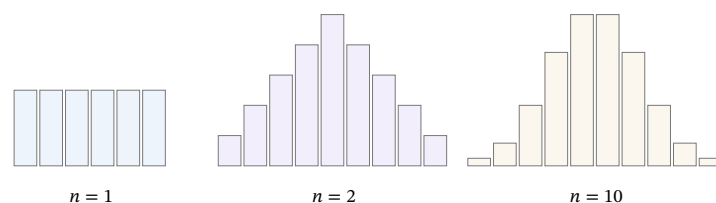


Figure 7: The Central Limit Theorem building a bell from a flat die. The distribution of the **sum** S_n of n dice: flat for $n = 1$, triangular for $n = 2$, and visibly bell-shaped by $n = 10$. The die's flat shape is irrelevant — only its mean and variance survive. The raw sum spreads without bound as n grows; it is the **standardized** sum $(S_n - n\mu)/(\sigma\sqrt{n})$ that settles onto the fixed $\mathcal{N}(0, 1)$ the theorem names.

5.2. Normal approximation of the binomial, with continuity correction

The first and most useful application: a binomial $X \sim \text{Bin}(n, p)$ is a sum of n i.i.d. Bernoulli(p) variables, so by the CLT it is approximately normal for large n :

$$\text{Bin}(n, p) \approx \mathcal{N}(np, np(1 - p)).$$

This rescues us from summing dozens of binomial terms by hand. One subtlety: we are approximating a **discrete** variable (integer counts) by a **continuous** one, so we spread each integer k across the interval $[k - 0.5, k + 0.5]$ — the **continuity correction**.

Example — at least 60 heads in 100 flips. Let $X \sim \text{Bin}(100, 0.5)$, so $np = 50$ and $\sqrt{np(1 - p)} = \sqrt{25} = 5$. For $\mathbb{P}(X \geq 60)$, the continuity correction replaces the cutoff 60 by 59.5:

$$\mathbb{P}(X \geq 60) \approx \mathbb{P}\left(Z \geq \frac{59.5 - 50}{5}\right) = \mathbb{P}(Z \geq 1.9) = 1 - \Phi(1.9) \approx 0.0287.$$

Without the correction we would use 60, giving $z = 2.0$ and 0.0228 — a 20% relative error in the tail. The half-unit shift matters most precisely where we care most: in the tails.

Pitfall. The continuity correction direction depends on the inequality. For $\mathbb{P}(X \geq k)$ use $k - 0.5$ (include the whole bar for k); for $\mathbb{P}(X \leq k)$ use $k + 0.5$; for $\mathbb{P}(X = k)$ use the band $[k - 0.5, k + 0.5]$. Dropping the correction is acceptable only when n is very large; for moderate n it is a real source of error. As a rule of thumb the approximation is trustworthy when $np \geq 5$ **and** $n(1 - p) \geq 5$.

5.3. Choosing a sample size

Because the standard error is $\sigma/\sqrt{n}$, the CLT lets us **design** a study in advance: solve for the n that delivers a target precision. If we want the sample mean within ε of μ with 95% confidence, we need $1.96 \sigma/\sqrt{n} \leq \varepsilon$, i.e.

$$n \geq \left(1.96 \frac{\sigma}{\varepsilon}\right)^2.$$

This single inequality — precision improves with $\sqrt{n}$, so cost grows with the **square** of the precision demanded — governs the economics of every poll, survey and A/B test.

6. Monte Carlo methods

The CLT does more than describe sampling — it **powers a computational method**. When a probability or an expectation is too hard to compute analytically (no closed-form integral, a complicated multidimensional region), we **simulate**: draw many random samples, average, and let the LLN and CLT do the rest. This is the **Monte Carlo** method, a mainstay of modern quantitative science.

6.1. Estimating an expectation by simulation

Example — estimating π by throwing darts. Throw darts uniformly at the unit square $[0, 1]^2$. A dart lands inside the quarter-circle $x^2 + y^2 \leq 1$ with probability equal to that region's area, $\pi/4$. So if a fraction $\hat{p}$ of N darts land inside, then $4\hat{p}$ estimates π — no geometry, just counting. We implement exactly this in Section 6.4.

Definition 10 (Monte Carlo estimator). To estimate $\theta = \mathbb{E}(g(X))$, draw independent samples $X_1, \dots, X_N$ from the distribution of X and average:

$$\hat{\theta}_N = \frac{1}{N} \sum_{i=1}^N g(X_i).$$

By the LLN $\hat{\theta}_N \rightarrow \theta$, and by the CLT its error is approximately normal with standard error $\sigma_g/\sqrt{N}$, where $\sigma_g^2 = \mathbb{V}(g(X))$.

The estimator is unbiased ($\mathbb{E}(\hat{\theta}_N) = \theta$ by linearity) and its $1/\sqrt{N}$ error rate is the same $\sqrt{N}$ law as before: a Monte Carlo answer with one more digit of accuracy costs 100 times the computation. The method is general — it estimates **any** expectation, hence any probability (take g an indicator) and any integral (write it as an expectation).

6.2. The inverse transform method

To simulate, we must first **generate** samples from the target distribution. A computer supplies only $U \sim \text{Unif}(0, 1)$. How do we turn uniform noise into, say, an exponential variable? The key is the CDF.

Theorem 10 (Inverse transform method). Let F be a continuous, strictly increasing CDF. If $U \sim \text{Unif}(0, 1)$, then

$$X = F^{-1}(U)$$

has CDF F . Conversely, $F(X) \sim \text{Unif}(0, 1)$ for any continuous X .

Proof. $\mathbb{P}(X \leq x) = \mathbb{P}(F^{-1}(U) \leq x) = \mathbb{P}(U \leq F(x)) = F(x)$, using that F is increasing (so we may apply it to both sides) and that $\mathbb{P}(U \leq u) = u$ for a standard uniform. $\square$

The picture: feeding a uniform height u on the vertical axis through the inverse CDF reads off the x whose accumulated probability is u . Regions where F rises steeply (high density) catch more of the uniform input, so they receive more samples — exactly as they should.

Example — simulating an exponential. The exponential CDF is $F(x) = 1 - e^{-\lambda x}$. Solve $u = 1 - e^{-\lambda x}$ for x : $x = -\ln(1 - u)/\lambda$. So if $U \sim \text{Unif}(0, 1)$ then $X = -\ln(1 - U)/\lambda$ is $\text{Exp}(\lambda)$. (Since $1 - U$ is also uniform, $-\ln(U)/\lambda$ works just as well.) One logarithm turns a uniform into an exponential.

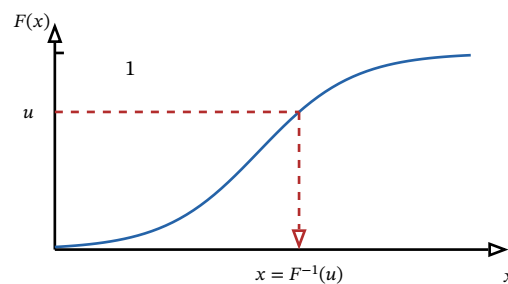


Figure 8: The inverse transform method. A uniform input $u \in (0, 1)$ on the vertical axis is mapped horizontally to the curve, then down to $x = F^{-1}(u)$. Steep parts of F (where the density is high) span more vertical room and so capture more uniform inputs — producing samples with the right density.

6.3. Confidence intervals for a Monte Carlo estimate

A simulation is worthless without an **error bar**. The CLT supplies one directly. Since $\hat{\theta}_N \approx \mathcal{N}(\theta, \sigma_g^2/N)$, and we estimate σ_g by the sample standard deviation s of the values $g(X_i)$, a 95% confidence interval for θ is

$$\hat{\theta}_N \pm 1.96 \cdot \frac{s}{\sqrt{N}}.$$

The half-width $1.96 s/\sqrt{N}$ is the **standard error band**: it shrinks like $1/\sqrt{N}$, so reporting it tells the reader exactly how much to trust the simulation. **A Monte Carlo result without this band is incomplete.**

6.4. The method in code

The recipe is short enough to write out in full. A single function estimates $\theta = \mathbb{E}(g(X))$ from N independent draws and returns three things at once: the estimate $\hat{\theta}_N$, its standard error $s/\sqrt{N}$, and the 95% confidence interval $\hat{\theta}_N \pm 1.96 s/\sqrt{N}$. Everything in this chapter fits in a dozen lines of Python — `sample` is a function that returns one draw of X , and g is the function whose expectation we want.

```
import random, math

def monte_carlo(sample, g, N):
    """Estimate theta = E[g(X)] from N draws: returns the estimate,
    its standard error, and a 95% confidence interval."""
    vals = [g(sample()) for _ in range(N)]
    mean = sum(vals) / N
    var = sum((v - mean) ** 2 for v in vals) / (N - 1) # sample variance
    se = math.sqrt(var / N) # standard error
    return mean, se, (mean - 1.96 * se, mean + 1.96 * se)
```

To estimate π , take g to be the indicator “the dart lands in the quarter-circle”: its expectation is exactly $\pi/4$, so multiplying the estimate by 4 estimates π — and the confidence interval scales by 4 with it.

```

random.seed(1)
def dart():
    # one uniform dart in the unit square
    return (random.random(), random.random())
def inside(point):
    # 1 if inside the quarter-circle, else 0
    x, y = point
    return 1.0 if x * x + y * y <= 1.0 else 0.0

p_hat, se, ci = monte_carlo(dart, inside, 100_000)
print(round(4 * p_hat, 4), (round(4 * ci[0], 4), round(4 * ci[1], 4)))

```

```
3.1378 (3.1276, 3.148)
```

The inverse transform supplies the sampler for any distribution. For the exponential, $F^{-1}(u) = -\ln(1 - u)/\lambda$, so one line turns `random()` into an $\text{Exp}(\lambda)$ draw; estimating $\mathbb{E}(X)$ with $g(x) = x$ should recover $1/\lambda$.

```

lam = 0.5
def exp_sample():
    # inverse transform: Unif(0,1) -> Exp(lambda)
    u = random.random()
    return -math.log(1 - u) / lam

mean, se, ci = monte_carlo(exp_sample, lambda x: x, 100_000)
print(round(mean, 4), (round(ci[0], 4), round(ci[1], 4)))

```

```
1.9951 (1.9828, 2.0075)
```

Both confidence intervals straddle the truth — $\pi = 3.14159\dots$ and $1/\lambda = 2$ — and both have half-width ≈ 0.01 at $N = 10^5$, exactly the $1/\sqrt{N}$ law in action. Shrinking that half-width to 0.001 would demand $N = 10^7$ draws: a further digit of accuracy costs a hundredfold more computation.

7. Hypothesis testing

We can now do **confirmatory** statistics: not merely estimate a quantity, but **test a claim** about it and quantify the evidence. Hypothesis testing is the disciplined procedure for deciding whether data are **surprising enough** under a default assumption to warrant rejecting it. The normal distribution, finally in hand, is what makes the arithmetic possible.

7.1. The logic: H_0 , H_1 , and a test statistic

Example — is the coin fair? A coin is flipped 100 times and lands heads 62 times. Is it biased, or is 62 just luck? We adopt a **sceptical default** — “the coin is fair, $p = 0.5$ ” — and ask: **if that were true, how surprising is an outcome as extreme as 62?** Under fairness $X \sim \text{Bin}(100, 0.5) \approx \mathcal{N}(50, 25)$. Because we are approximating a **discrete** count by the normal, we apply the continuity correction of Chapter 5: an outcome “as extreme as 62” starts at the lower edge 61.5 of the bar for 62, so the z-score is $(61.5 - 50)/5 = 2.3$. A result 2.3 standard errors out is rare — strong reason to doubt the default.

Definition 11 (null and alternative hypotheses). The **null hypothesis** H_0 is the default, no-effect claim we try to disprove (here $p = 0.5$). The **alternative hypothesis** H_1 is what we entertain if H_0 fails (here $p \neq 0.5$). A **test statistic** is a number computed from the data whose distribution **under** H_0 is known; we reject H_0 when the statistic falls in a **rejection region** of values too extreme to attribute to chance.

The asymmetry is deliberate and important: we never **prove** H_0 , we only **fail to reject** it. The burden of proof lies on the alternative, exactly as a court presumes innocence until evidence is strong enough to overturn it.

7.2. Significance level and the p -value

Definition 12 (significance level and rejection region). The **significance level** α (commonly 0.05) is the probability of rejecting H_0 **when it is actually true** — the false-alarm rate we are willing to tolerate. The **rejection region** is the set of test-statistic values whose total probability under H_0 equals α . For a two-sided z-test at $\alpha = 0.05$ it is $|z| > 1.96$.

Definition 13 (p -value). The **p -value** is the probability, **computed assuming H_0 is true**, of obtaining a result **at least as extreme** as the one observed. We reject H_0 exactly when $p \leq \alpha$.

For the coin, the two-sided p -value is $\mathbb{P}(|Z| \geq 2.3) = 2(1 - \Phi(2.3)) \approx 2(0.0107) = 0.0214$ — reassuringly close to the exact binomial value 0.021. Since $0.0214 < 0.05$, we reject fairness at the 5% level: an outcome this extreme would happen only about 2% of the time with a fair coin.

Pitfall. The p -value is the **probability of data this extreme given H_0** — written $\mathbb{P}(\text{data} \mid H_0)$. It is **not** the probability that H_0 is true, $\mathbb{P}(H_0 \mid \text{data})$. These are the two sides of Bayes’ theorem,

and confusing them is the most damaging error in applied statistics. “ $p = 0.02$ ” does **not** mean “there is a 2% chance the coin is fair”; it means “**if** the coin were fair, data this extreme would arise 2% of the time”. A small p -value casts doubt on H_0 ; it does not measure H_0 ’s probability.

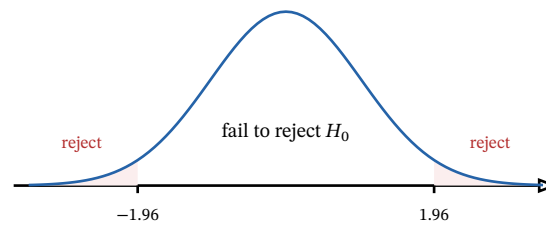


Figure 9: A two-sided z -test at $\alpha = 0.05$. Under H_0 the test statistic is standard normal; the two red tails beyond ± 1.96 together hold 5% of the mass. A statistic landing there is “too extreme to be chance”, so H_0 is rejected. The p -value is the area beyond the **observed** statistic.

7.3. One- and two-sided alternatives

Whether we look at one tail or two is decided by the **alternative** H_1 , which must be fixed **before** the data arrive. A **two-sided** alternative $H_1 : \mu \neq \mu_0$ asks “is the parameter different?”, and its p -value gathers **both** tails. A **one-sided** alternative $H_1 : \mu > \mu_0$ (or $\mu < \mu_0$) asks a directional question and gathers only the **one** tail in that direction.

Example — one tail or two?. A coaching course claims to **raise** scores above the national mean $\mu_0 = 100$ (known $\sigma = 15$). A class of $n = 50$ averages $\bar{x} = 104$, so the standard error is $15/\sqrt{50} = 2.12$ and $z = (104 - 100)/2.12 = 1.89$. For the directional claim “coaching helps”, $H_1 : \mu > 100$, the one-sided p -value is $\mathbb{P}(Z \geq 1.89) = 1 - \Phi(1.89) = 0.0294 < 0.05$ — significant. But the neutral question “does coaching change scores?”, $H_1 : \mu \neq 100$, doubles it to $2(0.0294) = 0.0588 > 0.05$ — **not** significant. Same data, opposite verdicts.

Pitfall. The direction of a one-sided test must come from the **science**, chosen before the data. Picking the side **after** seeing which tail the data fell in, then halving the p -value, secretly doubles the false-alarm rate — a form of p -hacking. When in doubt use the two-sided test: it is the honest default and never rewards peeking.

7.4. Choosing the statistic: z -test or t -test

Every test so far assumed a known population σ , which made the standardized statistic a **standard normal** — a z -test. In practice σ is almost always unknown and estimated from the sample as the sample SD s . Replacing σ by s injects a second source of uncertainty, and the statistic

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

then follows not the normal law but the **Student- t** law of Chapter 3, with $n - 1$ degrees of freedom. The choice is simple.

Framework 1 (z or t?).

- σ **known** (or n large, so $s \approx \sigma$): use the **z-test**, with critical values from the standard normal table of Chapter 3.
- σ **unknown and n small**: use the **t-test** with $n - 1$ degrees of freedom, critical values from the t -table below.

As $n \rightarrow \infty$ the two agree — t tends to the normal — so for large samples the distinction makes no practical difference.

The standard normal table of Chapter 3 serves the z-test; the t -test needs its own table of critical values, one row per number of degrees of freedom. The table below lists $t_{\alpha, k}$ — the value a Student- t with k degrees of freedom exceeds with probability α , i.e. its **upper-tail** critical value. Read it by the tail you need:

- a **one-sided** test at level α uses the column headed α ;
- a **two-sided** test at level α , and a confidence interval at level $1 - \alpha$ (Chapter 8), use the column headed $\alpha/2$ — half the rejected mass in each tail.

So a two-sided test at $\alpha = 0.05$ on 9 degrees of freedom reads row 9, column .025, giving $t_{0.025, 9} = 2.262$; the same column at $k = \infty$ returns the normal value 1.960, because $t \rightarrow \mathcal{N}(0, 1)$ as the sample grows.

d.f. k	upper-tail probability α				
	.10	.05	.025	.01	.005
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.779
27	1.314	1.703	2.052	2.473	2.771
28	1.313	1.701	2.048	2.467	2.763
29	1.311	1.699	2.045	2.462	2.756
30	1.310	1.697	2.042	2.457	2.750
40	1.303	1.684	2.021	2.423	2.704
60	1.296	1.671	2.000	2.390	2.660

d.f. k	upper-tail probability α				
	.10	.05	.025	.01	.005
120	1.289	1.658	1.980	2.358	2.617
∞	1.282	1.645	1.960	2.326	2.576

Because the table is a grid of **critical values**, not a CDF, a t -test's p -value is usually **bracketed** rather than read off exactly: find where the observed $|t|$ falls between two columns of its row, and its one-sided p -value lies between those two column headings α (double them for a two-sided p -value). That is enough to decide any test and to report the p -value to the precision a verdict needs.

Pitfall. The columns are **one-tailed** areas. For a two-sided test at $\alpha = 0.05$ do not read the .05 column — that is the one-sided 5% value — but the .025 column, so that each tail carries 2.5% and the two together carry the 5% you are spending. Reading the wrong column is the t -table's exact analogue of the left-tail/right-tail slip warned about for Φ in Chapter 3.

Example — a t -test for a small sample. A production line's batteries averaged $\mu_0 = 44$ hours. A sample of $n = 11$ from a new line gives $\bar{x} = 41$ hours with sample SD $s = 4$; the population σ is unknown. Testing $H_0 : \mu = 44$ against $H_1 : \mu \neq 44$ at $\alpha = 0.05$,

$$t = \frac{41 - 44}{4/\sqrt{11}} = -\frac{3}{1.206} = -2.49,$$

on $n - 1 = 10$ degrees of freedom. The two-sided critical value is $t_{0.025, 10} = 2.228$ (from the t -table above). Since $|-2.49| > 2.228$, we reject H_0 : the new line's mean lifetime differs from 44 hours. To report a p -value, bracket $|t| = 2.49$ in row 10: it sits between $t_{0.025} = 2.228$ and $t_{0.01} = 2.764$, so the one-sided p lies between 0.01 and 0.025 and the two-sided p between 0.02 and 0.05 — consistent with rejecting at $\alpha = 0.05$. Had we carelessly used the normal cutoff 1.96, we would have overstated our confidence — the t -test's wider cutoff is the honest price of not knowing σ .

7.5. Comparing two groups

The commonest test in practice concerns not one mean but the **difference** between two — treatment versus control, variant A versus variant B. The logic is unchanged: build a statistic measuring the observed difference in standard-error units, then read its tail probability under H_0 : no difference.

Example — an A/B test on conversion. An online store trials two checkout designs. Of $n_A = 1000$ visitors to design A, 200 convert ($\hat{p}_A = 0.20$); of $n_B = 1000$ to design B, 250 convert ($\hat{p}_B = 0.25$). Is B genuinely better, or is the 5-point gap chance? Under $H_0 : p_A = p_B$ we **pool** the samples to estimate the common rate,

$$\hat{p} = \frac{200 + 250}{1000 + 1000} = 0.225,$$

and the standard error of the difference is

$$SE = \sqrt{\hat{p}(1 - \hat{p})\left(\frac{1}{n_A} + \frac{1}{n_B}\right)} = \sqrt{0.225 \cdot 0.775 \cdot \left(\frac{1}{1000} + \frac{1}{1000}\right)} = 0.0187.$$

The test statistic is

$$z = \frac{\hat{p}_B - \hat{p}_A}{SE} = \frac{0.05}{0.0187} = 2.68,$$

with two-sided p -value $2(1 - \Phi(2.68)) = 2(0.0037) = 0.0074 < 0.05$. We reject H_0 : design B converts at a genuinely higher rate. (For the directional launch question “is B better?”, the one-sided $p = 0.0037$ tells the same story more sharply.)

The same two-sample pattern — with s -based standard errors and a t -distribution — compares two **means** from small samples. What never changes is the shape of the reasoning: a difference, divided by its standard error, judged against a known reference law.

7.6. Type I and Type II errors, and power

Every test can err in two opposite ways, and the trade-off between them is the core design problem of testing.

Definition 14 (the two error types). A **Type I error** (false positive) is rejecting H_0 when it is true; its probability is the significance level α . A **Type II error** (false negative) is failing to reject H_0 when H_1 is true; its probability is denoted β . The **power** of the test is $1 - \beta$, the probability of **correctly** detecting a true effect.

	H_0 true	H_1 true
reject H_0	Type I error (α)	correct (power $1 - \beta$)
fail to reject	correct	Type II error (β)

The two errors pull against each other: lowering α (demanding stronger evidence to reject) shrinks false alarms but **raises** β , missing more real effects. The only way to reduce **both** at once is to gather more data — a larger n sharpens the sampling distributions and increases power at any fixed α . This is why power calculations decide sample sizes before an experiment is run.

Pitfall. “Fail to reject H_0 ” is **not** “accept H_0 ” or “prove no effect”. A non-significant result may simply mean the test lacked the power to detect a real but small effect — **absence of evidence is not evidence of absence**. Reporting the power, or a confidence interval, is what distinguishes “we showed there is no effect” from “our study was too small to tell”.

8. Confidence intervals

A hypothesis test answers a yes/no question; a **confidence interval** answers the more useful quantitative one: **what range of parameter values is plausible given the data?** Built from the same CLT, it carries its own precise — and famously misread — interpretation.

8.1. Construction from the CLT

Example — estimating average height. We measure $n = 100$ people and find sample mean $\bar{x} = 171$ cm, with known population SD $\sigma = 8$ cm. By the CLT $\bar{X} \approx \mathcal{N}(\mu, 8^2/100)$, so the standard error is $8/\sqrt{100} = 0.8$ cm. With 95% probability $\bar{X}$ lands within 1.96 standard errors of μ — and we can turn that around into a statement about μ .

Definition 15 (confidence interval for a mean). From a sample mean $\bar{x}$ with standard error $\sigma/\sqrt{n}$, a 95% **confidence interval** for μ is

$$\bar{x} \pm 1.96 \cdot \frac{\sigma}{\sqrt{n}}.$$

The half-width $E = 1.96 \sigma/\sqrt{n}$ is the **margin of error**. For a general confidence level $1 - \alpha$, replace 1.96 by the critical value $z_{\alpha/2}$ with $\Phi(z_{\alpha/2}) = 1 - \alpha/2$.

For the heights: $171 \pm 1.96(0.8) = 171 \pm 1.57$, i.e. $[169.4, 172.6]$ cm.

8.1.1. When σ is unknown: the t -interval

The construction above assumed a **known** population SD σ . In practice we almost never know it and must estimate it from the sample itself as the sample SD s . For a small sample that estimate is itself uncertain, so we widen the interval by replacing the normal critical value with the slightly larger **Student- t** value $t_{\alpha/2, n-1}$ of Chapter 3, carrying $n - 1$ **degrees of freedom**:

$$\bar{x} \pm t_{\alpha/2, n-1} \cdot \frac{s}{\sqrt{n}}.$$

The critical value $t_{\alpha/2, n-1}$ is read from the t -table of Chapter 7; a two-sided 95% interval uses its .025 column ($\alpha/2 = 0.025$). That column shows the penalty fading as data accumulate: $t_{0.025} = 2.571$ at 5 degrees of freedom, 2.228 at 10, 2.086 at 20 and 2.042 at 30, closing on the normal's exact 1.960 at ∞ — with enough data, not knowing σ costs nothing.

Example — a t -interval for a small sample. Ten measurements of a part give $\bar{x} = 20.3$ mm with sample SD $s = 1.8$ mm. With $n - 1 = 9$ degrees of freedom the critical value is $t_{0.025, 9} = 2.262$, so the 95% interval is

$$20.3 \pm 2.262 \cdot \frac{1.8}{\sqrt{10}} = 20.3 \pm 1.29 = [19.0, 21.6].$$

The normal value 1.96 would give the narrower ± 1.12 — overstating our precision by ignoring that σ was itself only estimated.

8.2. What “95% confident” actually means

Pitfall. The interval $[169.4, 172.6]$ does **not** mean “there is a 95% probability that μ lies in it”. Once computed, the interval either contains the fixed (but unknown) μ or it does not — there is no probability left. The 95% is a property of the **procedure**: if we repeated the whole sampling-and-computing process many times, about 95% of the intervals produced would capture μ . **Confidence is in the method, not in any single interval.** Saying “ $\mathbb{P}(\mu \in [169.4, 172.6]) = 0.95$ ” treats the fixed parameter as random — precisely the error this caveat exists to prevent.

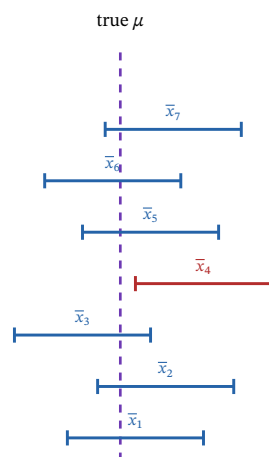


Figure 10: Seven samples, seven confidence intervals. Each brackets its own sample mean; the true μ (dashed line) is fixed. Most intervals capture μ , but one (red) misses entirely. Over many repetitions about 95% would capture it — that frequency **is** the confidence level. Any single interval simply does or does not.

8.3. Margin of error and sample size

The margin of error $E = z_{\alpha/2} \sigma / \sqrt{n}$ exposes the three levers of precision. A **narrower** interval demands a larger n (cost grows as E^{-2}), a smaller σ (less variable data), or a **lower** confidence level (accepting more misses). Inverting for n recovers the sample-size formula of Chapter 5:

$$n \geq \left(z_{\alpha/2} \frac{\sigma}{E} \right)^2.$$

8.4. Duality with hypothesis tests

Confidence intervals and two-sided tests are **two views of one computation**.

Proposition 4 (CI–test duality). A two-sided test of $H_0 : \mu = \mu_0$ at level α rejects H_0 **if and only if** μ_0 lies **outside** the $(1 - \alpha)$ confidence interval.

The interval is exactly the set of null values that the data would **not** reject — the “plausible” parameter values. This is often the better way to report a result: rather than a bare “reject / fail to reject”, the interval shows **which** values remain tenable and **how precisely** the parameter has been

pinned down. A test says whether μ_0 is plausible; the interval shows the whole range of plausible μ at once.

Example — reading the duality. For the heights, the 95% interval is $[169.4, 172.6]$. A claim “ $H_0 : \mu = 170$ ” would **not** be rejected at $\alpha = 0.05$, since 170 lies inside. A claim “ $H_0 : \mu = 168$ ” **would** be rejected, since 168 lies outside. We never recomputed a test — the interval already encodes every two-sided test at once.

Further reading

Everything in this book is standard, and two freely available texts develop it further with more examples and more depth. Read them alongside these chapters whenever you want a second exposition or extra practice.

- Jeremy Orloff and Jonathan Bloom, *Introduction to Probability and Statistics* (MIT OpenCourseWare 18.05, free): the closest match to this course. Its notes on continuous random variables, the central limit theorem, confidence intervals and significance testing cover exactly our chapters, with worked problems and solutions.
- Joseph K. Blitzstein and Jessica Hwang, *Introduction to Probability* (2nd edition, free PDF): a fuller reference. Chapter 5 (continuous random variables) and Chapter 10 (limit theorems) go deeper into the mathematics behind the Law of Large Numbers and the Central Limit Theorem.